Software Requirements Specification (SRS)

Full-Stack Cryptographic Blockchain Voting System

Project Title: Cryptographic Blockchain Voting System	**Status:** Approved
Document Version: 1.0.0	**Standard:** IEEE Std 830-1998
Target Audience: Engineers, Auditors, Cryptographers	**Date:** July 2026

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional, non-functional, security, and architectural requirements for the Full-Stack Cryptographic Blockchain Voting System. This document serves as the formal baseline for development, verification, and independent auditing.

1.2 Scope

The Blockchain Voting System is a production-grade digital voting platform engineered to guarantee end-to-end confidentiality, immutability, transparency, and anti-fraud detection. The system features:

- **AES-256 GCM Payload Encryption:** Ensures vote secrecy before transaction broadcast.
- **ECDSA SECP256R1 Signatures:** Provides mathematical proof of vote authenticity.
- **SHA-256 Merkle Tree Ledger:** Aggregates transactions into verifiable binary hash trees.
- **Proof-of-Work (PoW) Consensus:** Requires zero-prefix nonce calculation to prevent tampering.
- **AI Fraud Radar Module:** Computes dynamic threat risk scores (0-100%) based on velocity bursts.
- **QR Code Verification Receipts:** Enables voter auditability without revealing vote selection.

1.3 Definitions & Acronyms

Term / Acronym	Definition
AES-256 GCM	Advanced Encryption Standard with 256-bit key in Galois/Counter Mode.
ECDSA	Elliptic Curve Digital Signature Algorithm (SECP256R1 curve).
PoW	Proof-of-Work consensus algorithm requiring nonce discovery.
Merkle Tree	Binary hash tree for efficient transaction payload integrity verification.
RBAC	Role-Based Access Control (Admin, Voter, Observer).
JWT	JSON Web Token standard for stateless session authorization.

2. Overall Description

2.1 Product Perspective

The application operates as a decoupled full-stack architecture comprising a React 19 single-page client, an asynchronous FastAPI backend server, a SQLite relational database, and an embedded custom Python cryptographic blockchain.

2.2 User Classes & Characteristics

1. **Admin:** Manages election lifecycles, candidates, voter CSV rosters, audit logs, and AI risk metrics.
2. **Voter:** Views active elections, submits encrypted 1-click votes, receives QR code receipts, and verifies block inclusion.
3. **Observer:** Inspects public blockchain explorer, Merkle roots, live WebSockets tallies, and block signatures.

3. Specific System Requirements

3.1 Functional Requirements

Req ID	Module	Specification Description
FR-AUTH-01	Authentication	Supports login via username or email with bcrypt password verification.
FR-AUTH-03	Authentication	Issues short-lived access JWT and refresh JWT tokens for stateless RBAC.
FR-ELEC-01	Elections	Admin CRUD operations and state transitions (Draft -> Active -> Completed).
FR-VOTE-01	Voting Engine	Strict 1-vote constraint per user per election backed by database indices.
FR-VOTE-02	Voting Engine	AES-256 GCM encryption of vote selection payload before transaction queueing.
FR-VOTE-03	Voting Engine	ECDSA SECP256R1 digital keypair generation and transaction signing per vote.
FR-BC-02	Blockchain	Proof-of-Work mining with difficulty zero-prefix hash validation.
FR-BC-03	Blockchain	Computes and verifies binary Merkle Tree Root for all transactions in a block.
FR-RCPT-01	Receipts	Generates downloadable PDF/PNG QR Receipts with receipt SHA-256 hash.
FR-AI-01	AI Fraud Radar	Detects velocity bursts (<2s apart) and IP address concentration (>5 votes).
FR-AI-04	AI Fraud Radar	Computes unified threat score (0-100%) categorized as Low/Med/High/Critical.

3.2 Non-Functional Requirements (NFR)

- **NFR-SEC-01:** Zero plaintext candidate selections or user identities in mined blocks.
- **NFR-PERF-01:** API endpoint response latencies under 200ms.
- **NFR-PERF-02:** Proof-of-Work mining completes in 1.0 - 3.0s (difficulty = 2).
- **NFR-PERF-03:** Real-time WebSockets tally broadcast latency under 100ms.
- **NFR-REL-01:** Block payload immutability; downstream chain fails validation on tampering.

4. Verification & Compliance Matrix

Req ID	Verification Method	Acceptance Criteria / Status
FR-AUTH-02	Pytest Unit Test	Passwords hashed via bcrypt (salt cost >= 12). Passed.
FR-VOTE-01	Integration Test	Duplicate vote rejected with HTTP 400 Bad Request. Passed.
FR-VOTE-02	Crypto Audit	Payload unreadable without 256-bit AES GCM key. Passed.
FR-BC-05	Ledger Audit Endpoint	is_chain_valid returns true with 0 anomalies. Passed.